

Sun VME SCSI Theory of Operation

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ABSTRACT

This manual describes the circuitry of the Sun Microsystems VME SCSI board. The SCSI board is a dual-height VME board which contains an interface to the ANSI standard Small Computer Systems Interface (SCSI) bus, and a Real Time Clock circuit with battery backup.

1. Manual Overview

The SCSI board will be described in 3 sections: the SCSI section, the real time clock section, and the VME bus interface circuitry which is shared by the other two sections.

Definitions and Conventions

Host Bus	The VME bus or the internal bus which is just a buffered version of the VME bus. The VME bus signals names are prefixed by "P1." (e.g. - P1.D06), while the internal bus signal names are not (e.g. - D06).
Asserted	An active-low signal is asserted when it is low. An active-high signal is asserted when it is high.
Negated	A signal is negated when it is not asserted.
Signal Names	Active high signals have names which do not end with a minus sign (-) (e.g. - A03). Active-low signals have names which end with a minus sign (-) (e.g. - SEL-). Signals which are related to the SCSI circuitry only generally are prefixed with "S." (e.g. - S.DMAREQ). Signals which directly connect to one of the connectors along the top of the board end with square brackets ([]).

Component Designators

IC's are numbered Uxxx, where xxx is a number. The hundreds digit of the number tells which page of the schematic the part is on. If the part has multiple independent sections, sections are numbered starting at 0. In the case of 74F74 flip-flops, section 0 is the section whose clock input is pin 3, and section 1 is the one whose clock is pin 11. In general, the lower the pin numbers, the lower the section number.

2. SCSI Interface

The SCSI bus is an ANSI standard interface bus which is capable of connecting processors to peripherals and to other processors. It is intended to provide an interface protocol which is largely independent of the manufacturer of the peripheral device attached. SCSI peripheral devices may include disk drives, tape drives, printers, network interfaces, and others. A *Host Adapter* is needed to interface between the SCSI bus and the processor which wants to use the SCSI peripherals. The Sun SCSI Host Adapter comes in two versions: as a Multibus card and as a VME Bus card. The SCSI-specific circuits for the two versions are identical, but the bus interface circuits are obviously different. For further information about the SCSI bus, consult the *SCSI Buyers' Guide*, published by ADES. The *Controller/System Interface Specification* by Adaptec, Inc is also useful.

The Sun SCSI Host Adapter conforms to the SCSI spec. Arbitration is not supported. DMA is used for the data transfer phase of an SCSI transaction. Programmed I/O is used for the command, status, and message phases.

*****IMPORTANT***** This manual assumes that you understand the SCSI spec. If you don't, you may need to first read one of the documents referenced above.

The SCSI interface has 2 main sections: the SCSI bus interface and the DMA controller. The SCSI bus interface is the same for both the VME SCSI board and the Multibus SCSI board. The DMA controller is similar but not identical.

2.1. SCSI Bus Interface

The SCSI Bus Interface consists of the Data Register, the Command/Status Register, the Interface Control Register, the Parity circuit, and some control circuitry.

2.1.1. Register Addresses

Address	Size	Register
base+0x00	16	Data (read/write)
base+0x02	8	Command (write)/Status (read)
base+0x04	16	Interface Control (read/write)
base+0x06		reserved
base+0x08	16*	DMA Address High Word (read/write)
base+0x0a	16	DMA Address Low Word (read/write)
base+0x0c	16	DMA Count (read/write)
base+0x0e	8	Interrupt Vector Register (read/write)

* The DMA address register is 24-bits, thus only the low 8 bits of the DMA Address High Word are significant.

2.1.2. SCSI Connector

The SCSI bus connector is P2, a 96-pin DIN connector. The pinout is such that a 50-pin ribbon cable may be mounted on a ribbon-cable-mount DIN connector, and the signals will appear on the ribbon cable in their proper positions according to the SCSI spec. Consult the SCSI spec for the signal assignments on the ribbon cable, and page 9 of the VME SCSI schematic for the pin assignments on the DIN connector.

2.1.3. Interface Control Register

The Interface Control Register is used to control the SCSI host adapter by enabling/disabling various modes and by asserting or testing control lines on the SCSI bus. It consists of U100, U104, and U105. When the CPU writes to the Interface Control Register, the data present on the lower half of the host data bus will be latched into U100. This latch is automatically set to all zeroes whenever the SYSRESET- signal is asserted. Bits 4 and 5 connected to control lines on the SCSI bus via open-collector drivers in U108. If a 1 is written to either of these bits, the corresponding SCSI control line will be asserted (active low). Other bits in the lower byte control various modes of the Host Adapter, as described in the Sun SCSI Programmers Manual.

When the CPU reads the Interface Control Register, buffers U104 and U105 drive the host data bus. Some of the bits reflect the state of the host adapter. Others sense the SCSI control lines. U101 receives the SCSI control lines, providing hysteresis for noise immunity and inverting the active-low SCSI lines to the active-high logic convention used on the host adapter.

2.1.4. Data Register

The Data Register is used for transferring data from the host bus to the SCSI bus during the Data Phase of an SCSI transaction. It is 16 bits wide on the host bus side and 8 bits wide on the SCSI bus side. When it is read or written from the CPU, all 16 bits are accessed. When it is accessed from the SCSI side, one byte at a time is written, since the SCSI data bus is 8 bits wide. For information about how bytes are sequenced into and out of the Data Register, see the Byte Sequencing section elsewhere in this manual.

In addition, the Data Register is used to hold the selection ID's during the Selection Phase. The ID bits of the Host Adapter and the desired Target are written into the upper byte of the Data Register at the start of the selection procedure. The Data Register is used for this because during the Selection Phase the SCSI I/O, C/D, and MSG lines are in the same states as they are in for a "Data Out" transfer. Decoding is simplified by using the Data Register to hold the selection ID's, because the hardware need not treat the selection condition as being different from the Data Out condition. It also makes it possible to cope with controllers which either require the host adapter's ID to bit to be asserted (e.g. - the ADES Series 1 Disk Link) or do not allow the host adapter's ID bit to be asserted (contrary to the SCSI spec, e.g. - the Sysgen SC4000).

Note that due to the use of Am2952 bidirectional latches (U202, U203) for the Data Register, it is not possible to read back what has been written to the Data Register.

2.1.5. Command/Status Register

The Command/Status Register is used to send Commands and Messages and to receive Status and Messages. It is only 8 bits wide, appearing on the upper half of the host data bus. This register may only be accessed with programmed I/O. For writing, the data on the upper half of the host data bus will be latched into the register on the trailing edge of the write strobe. A byte that is written to this register will appear on the SCSI data bus when the SCSI C/D line is active (Command) and the SCSI I/O line is inactive (Output).

For transfers from the SCSI bus to the CPU (Status or Message In), the data on the SCSI data bus is latched into Command/Status Register on the leading edge of the SCSI REQ line, so long as the SCSI C/D line is active (Command) and the SCSI I/O line is active (Input). As for the Data Register, it is not possible to read back a value that has been written to the Command/Status Register.

2.1.6. Parity

If the Parity Enable bit in the Interface Control Register is set, U205 will generate and check odd parity on the SCSI data bus. During output operations, parity is asserted onto the SCSI parity line through an open-collector driver in U108. During input, the parity of the SCSI bus will be checked. If a parity error occurs on input, it will be latched by the "SCSI" control PAL, U102. The latching action is accomplished by internal feedback within the PAL. The indication is visible as bit 17 of the Interface Control Register. The only way to clear the indication is by momentarily clearing the Parity Enable bit.

2.1.7. SCSI Control Circuitry

The control circuitry of the SCSI Bus Interface performs these functions: SCSI Request sensing, SCSI Acknowledge generation, DMA Request generation, Interrupt generation, and Byte Sequencing. These functions interact with each other, and depend on the settings of two modes.

2.1.7.1. SCSI Modes

There are two modes which influence the behavior of the SCSI control logic. DMA Enable controls whether or not data transfers are done with DMA. Word Mode, when set, causes SCSI data bytes to be packed/unpacked in the Data Register, so that transfers over the host bus can transfer 2 bytes at a time. If Word Mode is off, a separate host bus transfer must occur for each byte that is transferred over the SCSI bus. The normal case is to turn on both DMA Enable and

Word Mode.

When Word Mode is on, a flip-flop (section 2 of U107) keeps track of whether the last byte that was transferred over the SCSI bus was an odd numbered byte or an even-numbered one. The output of this flip-flop is called SecondByte. SecondByte is initially off, and changes state after each data byte is transferred over the SCSI bus.

2.1.7.2. Byte Sequencing

In order to take advantage of the full width of the host data bus, 2 bytes from the SCSI data bus are packed into the Data Register. The 2 bytes are transferred over the host data bus in one operation, but only one byte at a time over the SCSI bus. The control section takes care of sequencing the bytes so that they go on the SCSI bus in the proper order.

For incoming data, the first byte goes into the upper half of the Data Register, and the second onto the lower half. At this point the Data Register is full and must be read from the host side before more transfers can take place. The SCSI REQuest is not acknowledged until the host has read the Data Register (either with DMA or programmed I/O) and thus emptied it. The cycle then repeats.

For outgoing data, the first request finds the Data Register empty, so the Data Register must be written with data from the host before the request may be acknowledged. When the second request comes in, it may be satisfied immediately, since there is still one byte in the lower half of the Data Register which hasn't been sent.

If Word Mode is off, the byte sequencing is turned off so that each individual request from the SCSI bus must be satisfied with a separate host access to the Data Register.

2.1.7.3. SCSI Request sensing

When the SCSI REQ line is asserted, the control circuitry decides whether or not the New Request bit in the Interface Control Register should be set. In general, New Request will be set only when the CPU must intervene to accomplish the transfer. It will not be set if the byte sequencing circuit or the DMA circuit can satisfy the REQuest without involving a programmed I/O operation.

2.1.7.4. SCSI Acknowledge generation.

Each byte transferred over the SCSI bus is accompanied by a REquest/ Acknowledge handshake. The SCSI Target device (controller) asserts REQuest, the host adapter either accepts or provides a data byte, the host adapter asserts ACKnowledge, the Target negates REQuest, and the host adapter negates ACKnowledge. The Sun host adapter asserts ACKnowledge whenever either the Data Register or the Command/Status Register is accessed from the host bus. The access may come from either a DMA operation or a programmed I/O operation in the case of the Data Register. The Command/Status Register is only accessible with programmed I/O. When either register is accessed, S.REGACC will be asserted by the "Decode" PAL, U200. The trailing edge of S.REGACC is remembered by U107 section 1, whose output signal S.REGACCL causes the "SCSI Control" PAL U102 to assert ACKnowledge. The S.REGACCL latch is cleared when the Target negates REQuest.

In Word Mode, half of the Data bytes are transferred without any transfer over the host bus, and thus without any host access of the registers. In this case, the SCSI Control PAL generates an automatic acknowledge as soon as the REQuest comes in.

Command/ Data*	Dma Enable	Input/ Output*	Word Mode	Second Byte	New Request	Auto Acknowl.	DMA Request
1	x	x	x	x	Yes	no	no
0	0	0	0	x	Yes	no	no
0	0	0	1	0	Yes	no	no
0	0	0	1	1	no	Yes	no
0	0	1	0	x	Yes	no	no
0	0	1	1	0	no	Yes	no
0	0	1	1	1	Yes	no	no
0	1	0	0	x	no	no	Yes
0	1	0	1	0	no	no	Yes
0	1	0	1	1	no	Yes	no
0	1	1	0	x	no	no	Yes
0	1	1	1	0	no	Yes	no
0	1	1	1	1	no	no	Yes

2.1.7.5. Interrupt Requests

An interrupt is generated during REQuests for Status, Messages, or Data with DMA Enable Off. Data does not cause an interrupt if the byte sequencing circuit takes care of the request without a host access to the Data Register. A bus error (described in the DMA Control section) will also cause an interrupt. If one of these conditions exists, and the Interrupt Enable bit is on, the CPU will be interrupted at a level determined by Jumper blocks J600 and J601. As soon as the request is acknowledged, the interrupt request goes away. The SCSI Control PAL takes care of this.

2.2. DMA Control (Page 3)

The DMA controller generates 24-bit DMA addresses and timing strobes. A count register may be used to set a maximum DMA count and ensure that that count is not exceeded. DMA cycles that start but do not finish within a predetermined time cause a bus error interrupt.

2.2.1. DMA Address Register

The DMA Address Register (U302, U303, U304) is a 24-bit wide counter. It is written from the host bus in 2 parts - the higher 8 bits and the lower 16 bits. The DMA Address Register may be read from the host bus, via buffers U305, U306, U307. The internal address bus M.A00-M.A23 is always driven by the address register to the current DMA address. The DMA Address Register counts bytes. The control circuitry is responsible for incrementing the DMA Address Counter twice if a word is transferred. Circuit Detail: Incrementing the counters is done by driving the function select lines on the 'LS461 counters rather than by driving the Carry In to the lowest counter. The reason is that the propagation delay from Carry In to Carry Out is too long for reliable operation at 8 MHz, considering other circuit delays. Even so, it requires two clock cycles at 8 MHz for each increment of the Address Register.

2.2.2. DMA Count Register

The DMA Count Register is a 16-bit counter which may be both read and written from the host bus. It is implemented with U300 and U301. It is used both to determine how many bytes were transferred with DMA, and to enforce a maximum count. The DMA Count Register counts in tandem with the DMA Address Register. When one of them increments, the other one does too. The DMA Count Register generates the signal S.OVERRUN, which is just Carry Out, when the count reaches -1 (or 65535, depending on your point-of-view). This signal is used by the control circuit to disable further DMA cycles. If the DMA Count Register is loaded with - Max-Count - 1 (ones complement of MaxCount), then the DMA Count Register will read -1 when the

maximum count has been reached, and no more DMA bytes may be transferred.

2.2.3. DMA Cycle Generator (Pages 4 and 5)

The control circuit is responsible for running the DMA cycle in response to a DMA request. This involves detecting the DMA Request, arbitrating for the use of the host bus, generating the timing strobes, and incrementing the counters.

2.2.3.1. VME DMA control implementation

The DMA controller is implemented as a state machine clocked by an 8 MHz clock BCLK. BCLK is derived from the VME Bus P1.SYSCLK (16 Mhz) by the U404 section 0, which is connected as a divide-by-two counter.

2.2.3.2. Request Generation

A new DMA request is initiated by a rising edge on S.XREQ. If there is no bus error condition, this edge will set flip-flop U402 section 1, driving S.DMAREQ active. S.DMAREQ goes to the bus request PAL U501, which then asserts the appropriate bus request level through U500 section 2 and Jumper block J501.

2.2.3.3. Arbitration

Arbitration is performed by an external VME bus arbiter, which is on the CPU board in Sun VME systems. The external arbiter grants the bus by asserting P1.BGxIN- where x is the appropriate request level. Jumper block J500 must be jumpered to the same level as J501; J500 passes the bus grant signal to the U501 requestor pal, which causes a rising edge on EVENT (U501 pin 19). This rising edge clocks U502 section 0, which latches up the state of the S.DMAREQ line. D500 and U502 section 1 provide a delay to allow the output of U502 to settle. D500 provides the delay from the rising edge of EVENT. U502 turns the delayed signal off immediately on the falling edge of EVENT. When this settling delay has passed, U501 makes the decision about who gets the bus. Assuming that S.DMAREQ was high when U502 section 0 was clocked, U501 will give control of the VME bus to the SCSI board by asserting P1.BBSY- through U500 section 0, releasing P1.BRx-, and asserting S.DMAGR and S.DMAGR-.

Once the requestor has gained control of the bus in this fashion, it will retain control until another master requests the bus by asserting P1.BR0-, P1.BR1-, P1.BR2-, or P1.BR3-. Whenever one of these bus request lines is asserted, or when a later request from the SCSI board comes along (by another rising edge on S.DMAREQ), the EVENT signal goes high, reclocking U502 section 0. At a time that U502 section 0 is clocked, U501 will wait for the settling delay and then decide to either grant the bus to the SCSI board (if S.DMAREQ was high when the clock edge occurred), or to give the bus up (if S.DMAREQ was low).

2.2.3.4. Timing Generation

Once S.DMAGR is asserted, the DMA master machine runs a VME cycle. First, addresses are enabled onto the VME bus by S.DMAGR through U711, U712, and U713. At the same time, the VME data transceivers (U706, U707) are enabled by D.EN.L- and D.EN.U-, which are generated by U708 whenever S.DMAGR- is asserted. Next, S.DMAGR is synchronized to BCLK by U403 section 0, generating S.DMAGR.S. When U406 sees S.DMAGR.S go active, it runs a VME cycle.

If the cycle is a write to DMA memory, S.RD.DMA- is asserted as soon as S.DMAGR is asserted. This enables the SCSI Data Registers (U202, U203) onto the internal data bus. The enable signal first passes through U110 section 2. Next, U406 asserts EN.AS-, which tells U405 to generate P1.AS-, P1.DS0-, and P1.DS1- through U408. U405 decides which of the data strobes to assert by looking at M.A00 and S.EN.16BIT, as follows:

S.EN.16BIT M.A00 P1.DS0- P1.DS1-

1	x	ON	ON
0	0	OFF	ON
0	1	ON	OFF

For a read from DMA memory, the situation is similar, except that S.WR.DMA- is asserted instead of S.RD.DMA-, and S.WR.DMA- is asserted at the same time as EN.AS-, instead of before it. The reason that S.RD.DMA- is asserted for a memory write is that in order to write to the memory, you have to read the SCSI Data Register, and S.RD.DMA- goes to the SCSI Data Register, not to the VME bus. Similar comments apply for S.WR.DMA-.

In either case, the master pal U406 now waits for the cycle to be completed. The cycle can be completed in one of three ways.

- 1) The cycle can be terminated successfully by P1.DTACK-.
- 2) The cycle can be aborted by a VME bus error P1.BERR-.
- 3) The cycle can be aborted by an on-board timeout.

The P1.DTACK- and P1.BERR- signals pass through U701, are or-ed together in U405, and are synchronized to BCLK by U403. The on-board timeout is generated by an RC delay using U400 section 0, R400, C400, and U408.

Either kind of error will cause the ERROR signal to go active, clocking U402 section 0 so that it will generate S.BUSERR and prevent further DMA requests.

Once ACK.S has been asserted, U406 takes away EN.AS- and either S.RD.DMA- or S.WR.DMA-, ending the cycle as far as the VME bus is concerned. During the next cycle, the DMA counters are incremented once by S.INC-. One cycle later, S.BRCLR- is asserted, which clears the DMA request latch U402 section 1. On the final cycle, the DMA counters are again incremented if EN16BIT is asserted.

2.2.3.5. DMA Overrun

If the SCSI Target devices tries to transfer more bytes than the maximum number that the host software set up in the DMA count register, the hardware will prevent the extra bytes from causing DMA transfers. To set a maximum count, the software writes (-MaxCount - 1) to the DMA Count Register. When the maximum count has been reached, the DMA Count Register will have counted to -1, causing the carry out signal from the counters to be active. This signal is labelled S.OVERRUN. If exactly the maximum count is transferred, nothing bad will happen. However, if another transfer is attempted while the S.OVERRUN signal is asserted, U406 will not issue the EN.AS- signal, so the VME cycle won't really happen. Since the VME strobes won't be asserted, no slave device will respond to the DMA cycle, causing a timeout Bus Error as described above. The Bus Error will complete the DMA control PAL's cycle, incrementing the DMA count register past -1 to either 0 or 1, depending on whether Word Mode is on or off. No further DMA cycles will be requested, due to the Bus Error condition. The Overrun condition may be distinguished from a "normal" Bus Error by the fact that the DMA count register reads 0 or 1.

3. Real Time Clock (Page 8)

The Real Time Clock circuit is based on a National 58167 chip U803. This chip has very long setup, hold, and access times, so the data going to and from it needs to be latched. This is done by U802 and U804. Pal U801 controls these latches and generates the read and write strobes for the 58167. See the timing diagram for U801 for more information. U106 section 1 prevents the A4 line of the clock chip from toggling unless the clock chip is actually being accessed. This is needed because a design problem in the chip itself causes activity on the A4 line to couple into the oscillator circuit and affect the accuracy of the clock.

The oscillator circuit is made of X800, C801, C802, R811, and R813. R811 sets the power level at which the crystal operates. C801 is a trimmer capacitor which is used to adjust the precise frequency of the oscillator so it will keep good time. R813 gets around another design bug in the chip, providing bias current for the oscillator. Most of the 58167 chips can do without this resistor, but a few won't oscillate without it.

Battery backup for the real time clock chip is provided by battery BATT1, along with R812, D805, and C803. The battery is a long-life lithium type, so it doesn't need to be recharged.

When the main system 5V power is absent, the 58167 chip must be put in power-down mode. The power fail sensor circuit, U805 and associated components, perform this function. When the +5V supply is turning on, C800 delays the turn-on of the circuit until the supply has had time to stabilize. R804 and R805 provide fast turn-off when the 5V supply is going off.

4. Interrupts (Page 6)

Interrupt requests from the SCSI section cause VME.INTREQ to be asserted. This is driven onto the VME bus by U500 section 3, through jumper block J601, which select the level.

When the CPU acknowledges an interrupt, it will assert IACK, IACKIN, and AS. When AS and IACK are both true (U603 section 0), U600 section 0 will be clocked, sampling VME.INTREQ. U601 and J600 compare the interrupt acknowledge level with the SCSI board's interrupt level. If the interrupt acknowledge is for the level that the SCSI board is interrupting on, the signal on the right side of J600 will go low. D500 provides a delay to give U600 and U601 time to settle. Once the delay has expired, either U604 section 0 will assert I.DTACK- (it was our interrupt), or U604 section 1 will propagate P1.IACKOUT- (it was some other board's interrupt). If it was our interrupt, the Interrupt Vector Register U602 will be enabled onto the internal data bus.

4.1. Interrupt Level

SCSI normally interrupts at level 2, so jumpers should be installed at the level 2 positions of J600 and J601.

5. VME Bus Slave Interface and Drivers (Page 7)

5.1. Bus Drivers/Receivers

U711, U712, and U713 are address drivers. The VME Address Bus is driven only when S.DMAEN is active, which only occurs during a DMA cycle as described earlier. U706 and U707 are data transceivers. U701 buffers several of the VME bus control lines. U700 latches the state of the low-order address lines and the P1.LWORD- and P1.WRITE- signals so that glitches on these lines won't cause problems. High-order address lines are deglitched differently.

5.2. Decoding

The SCSI board responds to a 4 Kbyte bank of addresses, or 2 pages. The first 2 K page is used for the SCSI registers. The second 2 K is used for the real time clock circuit. The SCSI registers and the real time clock are on separate 2 K boundaries so that the memory protection hardware on the CPU can protect the devices separately (protection applies to 2 K pages). The base address for the SCSI board is set with Dip Switches U702 and U704. When the twelve high order address bits P1.A12 through P1.A23 match the address selected with U702 and U704, ADDRMATCH will be asserted. ADDRMATCH also depends on AS- being asserted, LWORD- being negated, and P1.AM3, P1.AM4, and P1.AM5 all being high. A switch on (closed) corresponds to an address bit negated (1 for the software or electrical high on the VME address bus).

U709 decodes ADDRMATCH, AS-, DS0, DS1, the rest of the address modifier lines P1.AM0, P1.AM1, P1.AM2, P1.ACFAIL-, P1.IACK-, P1.A11, SYSRESET, and S.DMAGR-. It

generates 4 signals RTC.SEL-, SCSI.SEL-, BOARDSEL-, and IACK. SCSI.SEL- is asserted on a slave access when P1.A11 is low. RTC.SEL- is asserted on a slave access when P1.A11 is high. BOARDENA- is asserted whenever either of SCSI.SEL- or RTC.SEL- is asserted. IACK is only asserted on an interrupt acknowledge cycle. Deglitching is provided by self-latching in the pal; whenever either of SCSI.SEL- or RTC.SEL- is activated, it latches itself on for the remainder of the cycle.

5.3. Data Transfer Acknowledge (DTACK) (Page 7)

Slave access cycles and Interrupt Acknowledge cycles all require that P1.DTACK- be generated. In the case of an SCSI access, the registers can be accessed in a very short time, less than 125 nsecs. Interrupt acknowledge cycles can be completed quickly too. The real time clock takes much longer, about 1 microsecond. The time delay for the real time clock is provided by the PAL that controls the clock chip, U801. SCSI.SEL-, I.DTACK-, and R.DTACK- are OR-ed together U708 and fed to U710. U710 provides a little extra delay, between 1 and 2 clock periods, to guarantee enough time for SCSI cycles and interrupt acknowledge cycles. After this delay, U400 drives the P1.DTACK- signal onto the bus. The delayed signal, X.DTACK-, is also used to gate off the write strobes to the SCSI registers so that the hold time will be satisfied (see U901).

5.4. Bus Priority

The Sun VME systems normally use Bus Priority Level 3 for everything, so jumpers should be installed for level 3 in J500, J501, and J502.